

Vimal Jyothi Engineering College
(Autonomous)
DEPARTMENT OF COMPUTER SCIENCE ENGINEERING
Vth Semester - B.Tech
Course : PCCST501 - COMPUTER NETWORKS

QNo	QUESTION Module 4 Submodule 1 Topic 1	CO	MARK
1.	Define Hamming distance and minimum Hamming distance. Calculate the pairwise Hamming distance and minimum Hamming distance among the following code words: 100000, 100110, 111101. (Problem) (Applying(P)) (Medium) (Ref : Question Paper) (B Tech Degree S5 (S, FE) /S3 (PT) (S) Examination June 2024 (2019 Scheme))	CO4	(9.00)
2.	Explain single-error correction using Hamming code with a suitable example. Construct the Hamming code word for the data 1001101. (Problem) (Applying(P)) (Medium) (Ref : Book) (Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Chapter 10 – Error Detection and Correction (Hamming Code))	CO4	(9.00)
3.	A bit stream 10011101 is transmitted using the CRC method. The generator polynomial is $x^3 + 1$. Show the actual bit string transmitted. (Problem) (Applying(P)) (Medium) (Ref : Question Paper) (B.Tech Degree 55 (R, s) / 53 (PT) (R, S) Examination Decdmber 2023 (2019 Scheme))	CO4	(9.00)
4.	The following character encoding is used in a data link protocol: A = 01000111 B = 11100011 FLAG = 01111110 ESC = 11100000 Show the bit sequence transmitted (in binary) for the four-character frame A B ESC FLAG when each of the following framing methods are used: (Problem) (Applying(P)) (Medium) (Ref : Book) (Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Chapter 11 – Data Link Control (Framing: Character Count, Byte Stuffing, Bit Stuffing).)	CO4	(9.00)
5.	A bit stream 1010001101 is transmitted using the standard CRC method. The generator polynomial used is 110101. Show the actual bit string transmitted. (Problem) (Applying(P)) (Medium) (Ref : Book) (Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Chapter 10 – Error Detection and Correction (CRC))	CO4	(9.00)
6.	Explain the significance of Hamming distance in error detection and correction. What is the hamming distance between the code word ‘11001011’ and ‘10000111’? (Problem) (Applying(P)) (Medium) (Ref : Question Paper) (B.Tech Degree S5 (R,S) (FT/WP) (S3 PT) Examination November 2025 (2019 Scheme).)	CO4	(3.00)
7.	Differentiate between Asynchronous Response Mode (ARM) and Asynchronous Balanced Mode (ABM) of operation in HDLC. (Theory) (Understanding(U)) (Medium) (Ref : Question Paper) (B Tech Degree S5 (S, FE) /S3 (FT/WP)(S3 PT) Examination May 2025 (2019 Scheme))	CO4	(3.00)

8. A bit string 0111101110111110 needs to be transmitted at the data link layer. If the flag used is 01111110, what is the string actually transmitted after bit stuffing?
(Problem) (Applying(P)) (Medium) (Ref : Question Paper) (B Tech Degree S5 (R,S) Examination 2021 (2019 Scheme))

Submodule 2
Topic 1

9. Write short notes on any two Carrier Sense Multiple Access (CSMA) protocols used in IEEE standards.
(Theory) (Understanding(U)) (Medium) (Ref : Question Paper) (B.Tech Degree S5 (R,S) Examination, December 2022 (2019 Scheme))
10. Differentiate between contention-based protocols and collision-free protocols for medium access. Explain two protocols from each type.
(Theory) (Understanding(U)) (Medium) (Ref : Question Paper) (B Tech Degree S5 (S, FE) /S3 (FT/WP)(S3 PT) Examination May 2025 (2019 Scheme))
11. Discuss about the frame formats of HDLC protocol.
(Theory) (Understanding(U)) (Medium) (Ref : Question Paper) (B.Tech Degree S5 (R,S) Examination, December 2022 (2019 Scheme))
12. Differentiate between 1-persistent and p-persistent CSMA.
(Theory) (Understanding(U)) (Medium) (Ref : Question Paper) (B.Tech Degree S5 (R,S) Examination, December 2023 (2019 Scheme))
13. How does a switch use MAC addresses to forward frames?
(Theory) (Understanding(U)) (Medium) (Ref : Book) (Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Chapter 15 – Connecting Devices (Switches/Layer-2 Switches).)
14. Differentiate between pure Aloha and slotted Aloha.
(Theory) (Understanding(U)) (Medium) (Ref : Question Paper) (B Tech Degree S5 (S, FE) /S3 (PT)(S) Examination June 2024 (2019 Scheme))

Submodule 3
Topic 1

15. Explain the IEEE 802.3 Ethernet cabling standards and frame format with suitable diagrams.
(Theory) (Understanding(U)) (Medium) (Ref : Question Paper) (B.Tech Degree S5 (R, S) / 55 (WP) (R) / 53 (PT) (S,FE) Examination November 2024 (2019 Scheme))
16. Which are the devices operating in the Data link layer? Explain the function of each of them.
(Theory) (Understanding(U)) (Medium) (Ref : Question Paper) (B.Tech Degree S5 (R,S) Examination, December 2021 (2019 Scheme))
17. Compare 10Base2, 10Base5, 10Base-T and 10Base-F Ethernet cabling.
(Theory) (Understanding(U)) (Medium) (Ref : Question Paper) (B Tech Degree S5 (S, FE) /S3 (PT)(S) Examination June 2024 (2019 Scheme))
18. Draw the frame format of an Ethernet frame and explain the function of each field.
(Theory) (Understanding(U)) (Medium) (Ref : Question Paper) (B Tech Degree S5 (S, FE) /S3 (PT)(S) Examination June 2024 (2019 Scheme))
19. Differentiate between bridges and switches.

(Theory) (Understanding(U)) (Easy) (Ref : Question Paper) (B Tech Degree S5 (R,S)
Examination 2022 (2019 Scheme))

20. Show how the data field is managed in the Ethernet frame format. CO4 (3.00)

(Theory) (Understanding(U)) (Medium) (Ref : Question Paper) (B Tech Degree S5 (R,S)
Examination 2022 (2019 Scheme))

21. List the features of Fast Ethernet. CO4 (3.00)

(Theory) (Remembering(R)) (Easy) (Ref : Question Paper) (B Tech Degree S5 (S, FE) /S3 (PT)
(S) Examination June 2024 (2019 Scheme))

Submodule 4

Topic 1

22. Explain the architecture and key features of 5G networks. Discuss the major technologies that enable 5G communication. CO4 (9.00)

(Theory) (Understanding(U)) (Medium) (Ref : Book) (Behrouz A. Forouzan, Data
Communications and Networking, 5th Edition)

23. Explain the hidden station problem and the exposed station problem in IEEE 802.11 Wireless LANs. How are these problems solved? CO4 (9.00)

(Theory) (Understanding(U)) (Medium) (Ref : Question Paper) (B Tech Degree S5 (S, FE) /S3
(FT/WP)(S3 PT) Examination May 2025 (2019 Scheme))

24. Draw and explain IEEE 802.11 Wireless LAN frame structure. CO4 (9.00)

(Theory) (Understanding(U)) (Medium) (Ref : Question Paper) (B.Tech Degree S5 (R,S)
Examination, December 2021 (2019 Scheme))

25. Define DCF and PCF in the context of wireless communication. CO4 (3.00)

(Theory) (Understanding(U)) (Easy) (Ref : Book) (Behrouz A. Forouzan, Data Communications
and Networking, 5th Edition, Chapter 16 – Wireless LANs (IEEE 802.11 MAC Sublayer).)

26. Explain the role of SSID in IEEE 802.11 wireless networks. CO4 (3.00)

(Theory) (Understanding(U)) (Easy) (Ref : Book) (Behrouz A. Forouzan, Data Communications
and Networking, 5th Edition, Chapter 16 – Wireless LANs (IEEE 802.11).)

27. Define RTS and CTS in the context of wireless communication. CO4 (3.00)

(Theory) (Remembering(R)) (Easy) (Ref : Question Paper) (Behrouz A. Forouzan, Data
Communications and Networking (5th Ed.), Chapter 16 – Wireless LANs)

Submodule 5

Topic 1

28. Explain the roles of the Home Agent and Foreign Agent in Mobile IP. Describe the three phases of Mobile IP communication with the help of a neat diagram. CO4 (9.00)

(Theory) (Understanding(U)) (Medium) (Ref : Book) (Behrouz A. Forouzan, Data
Communications and Networking, 5th Edition, Chapter 9 –Network Management(Mobile IP).)

29. Discuss the different data types used in ASN.1. Explain simple variables, structured types (SEQUENCE), and structured variables with appropriate examples. CO4 (9.00)

(Theory) (Understanding(U)) (Medium) (Ref : Book) (Behrouz A. Forouzan, Data
Communications and Networking, 5th Edition, Chapter 9 –Network Management(ASN.1).)

30. Explain how SNMP uses the Management Information Base (MIB) for network management. Can an SNMP message reference a leaf node in the MIB tree? Justify your answer with a suitable example. CO4 (9.00)

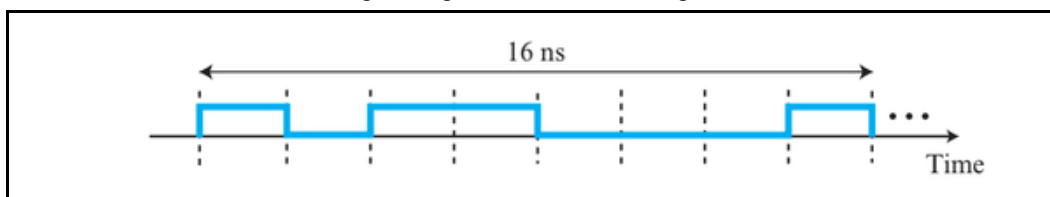
(Theory) (Understanding(U)) (Medium) (Ref : Book) (Behrouz A. Forouzan, Data
Communications and Networking, 5th Edition, Chapter 9 –Network Management(SNMP).)

- 31.** What are the three main components of Internet network management? State the role of each. CO4 (3.00)
(Theory) (Remembering(R)) (Easy) (Ref : Book) (Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Chapter 9 –Network Management(SNMP).)
- 32.** Explain the two address approach used in Mobile IP. CO4 (3.00)
(Theory) (Understanding(U)) (Easy) (Ref : Book) (Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Chapter 9 –Network Management(SNMP).)
- 33.** List and explain any three features or functions of SNMP. CO4 (3.00)
(Theory) (Understanding(U)) (Easy) (Ref : Book) (Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Chapter 9 –Physical Layer and Transmission Media (SNMP).)

Submodule 6

Topic 1

- 34.** Calculate the bit rate of the digital signal shown in the figure below. CO5 (3.00)



- (Problem) (Applying(P)) (Medium) (Ref : Book) (Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Chapter 7 –Physical Layer and Transmission Media (Data and Signals))
- 35.** Explain the different types of transmission impairments that affect signal quality during data transmission. Discuss attenuation in detail and explain how it is measured using the decibel (dB) formula. CO5 (9.00)
(Theory) (Understanding(U)) (Medium) (Ref : Book) (Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Chapter 7 –Physical Layer and Transmission Media (Data and Signals))
- 36.** What is the bandwidth of a signal that can be decomposed into five sine waves with frequencies at 0, 20, 50, 100, and 200 Hz? All peak amplitudes are the same. Draw the signal in the frequency domain and indicate the bandwidth. CO5 (9.00)
(Problem) (Applying(P)) (Medium) (Ref : Book) (Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Chapter 7 –Physical Layer and Transmission Media (Data and Signals))
- 37.** Explain the concept of bit rate in digital communication. Using the relation between bit duration and bit rate, calculate the bit rate for the following signals: CO5 (9.00)
(a) A signal in which 1 bit lasts 0.001 s
(b) A signal in which 1 bit lasts 2 ms
(c) A signal in which 10 bits last 20 μ s
Show all necessary calculations.
(Problem) (Applying(P)) (Medium) (Ref : Book) (Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Chapter 7 –Physical Layer and Transmission Media (Data and Signals).)
- 38.** How many bits can fit on a link with a 4 ms delay, if the bandwidth of the link is CO5 (3.00)
a. 1 Mbps?
b. 10 Mbps?
c. 100 Mbps?
(Problem) (Applying(P)) (Medium) (Ref : Question Paper) (B Tech Degree S5 (R, S) /S5(WP)/S3(PT)(S,FE) Examination November 2024 (2019 Scheme))
- 39.** A company needs to connect two computers using a dedicated cable. Which transmission technique (baseband or broadband) is more suitable? Justify your choice. CO5 (3.00)

(Theory) (Applying(P)) (Medium) (Ref : Book) (Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Chapter 7 –Physical Layer and Transmission Media (Analog and Digital).)

Submodule 7

Topic 1

- 40.** Illustrate the process of Analog-to-Analog Conversion and explain the working of AM, FM, and PM modulation techniques with suitable diagrams. CO5 (9.00)
- (Theory) (Understanding(U)) (Medium) (Ref : Question Paper) (Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Chapter 7 –Physical Layer and Transmission Media (Digital transmission, Analog Transmission).)
- 41.** Discuss the working principles of ASK and PSK digital-to-analog conversion mechanisms. Explain which analog signal characteristics are varied in each technique and illustrate them with neat diagrams. CO5 (9.00)
- (Theory) (Understanding(U)) (Medium) (Ref : Book) (Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Chapter 7 –Physical Layer and Transmission Media (Digital transmission, Analog Transmission).)
- 42.** Name two line coding schemes which provide self-synchronization. Justify your answer. Draw the patterns for the following data: 10101111, 11110000, 00010110 for those line coding schemes. CO5 (9.00)
- (Problem) (Applying(P)) (Medium) (Ref : Question Paper) (B.Tech Degree S5 (S,FE) (FT/WP), (S3 PT) Examination May 2025 (2019 Scheme))
- 43.** Draw the Binary encoding, Manchester encoding, and differential Manchester encoding of the following bit pattern: 0001110101. CO5 (9.00)
- (Problem) (Applying(P)) (Medium) (Ref : Question Paper) (B.Tech Degree S5 (R, S) / S5 (WP) (R) / S3 (PT) (S,FE) Examination November 2024 (2019 Scheme))
- 44.** Discuss the different signal encoding schemes used in the physical layer. Sketch waveforms for the binary sequence 10100111001 using Manchester and Differential Manchester encoding schemes. CO5 (9.00)
- (Problem) (Applying(P)) (Medium) (Ref : Question Paper) (B.Tech Degree S5 (R,S) (FT/WP) (S3 PT) Examination November 2025 (2019 Scheme).)
- 45.** Differentiate between ASK, FSK, and PSK based on the carrier parameter that is modified and the method of representing binary data. CO5 (3.00)
- (Theory) (Understanding(U)) (Medium) (Ref : Book) (Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Chapter 7 –Physical Layer and Transmission Media (Analog Transmission).)
- 46.** What are the different techniques used in digital-to-digital conversion? Briefly explain each. CO5 (3.00)
- (Theory) (Understanding(U)) (Easy) (Ref : Book) (Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Chapter 7 –Physical Layer and Transmission Media (Digital Transmission).)
- 47.** Encode the bit pattern 10110100110 using the Differential Manchester encoding technique. Sketch the corresponding waveform. CO5 (3.00)
- (Problem) (Applying(P)) (Medium) (Ref : Question Paper) (B.Tech Degree S5 (S, FE) / S3 (PT) (S) Examination June 2024 (2019 Scheme))

Submodule 8

- 48.** Compare Frequency Division Multiplexing (FDM), Wavelength Division Multiplexing (WDM), and Time Division Multiplexing (TDM) with respect to working principle, CO5 (9.00)

signal type, applications, and advantages.

(Theory) (Understanding(U)) (Medium) (Ref : Book) (Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Chapter 7 –Physical Layer and Transmission Media (Bandwidth Utilization).)

Topic 1

- 49.** Illustrate the construction of optical fiber and justify how this reduces the interference. CO5 (9.00)
Draw the structure of optical fiber.
(Theory) (Understanding(U)) (Medium) (Ref : Question Paper) (Fifth Semester B.Tech Degree Regular and Supplementary Examination December 2022 (2019 Scheme))
- 50.** List the different guided media for communication. Explain the cable type, connectors, and applications of each of them. CO5 (9.00)
(Theory) (Understanding(U)) (Easy) (Ref : Question Paper) (Fifth Semester B.Tech Degree Examination December 2021 (2019 Scheme))
- 51.** Discuss about guided and unguided media with appropriate diagram. CO5 (9.00)
(Theory) (Understanding(U)) (Medium) (Ref : Question Paper) (B.Tech Degree S5 (R,S) (FT/WP)(S3 PT) Examination November 2025 (2019 Scheme).)
- 52.** List the three major classes of guided transmission media and briefly explain each. CO5 (3.00)
(Theory) (Remembering(R)) (Easy) (Ref : Book) (Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Chapter 7 –Physical Layer and Transmission Media (Transmission Media).)
- 53.** A communication link operates over the frequency range of 20 kHz to 32 kHz. Using Frequency Division Multiplexing (FDM), allocate suitable frequency bands for three voice channels, each having a bandwidth of 4 kHz. Assume that no guard bands are used. Draw the frequency-domain configuration. CO5 (3.00)
(Problem) (Applying(P)) (Medium) (Ref : Question Paper) (Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Chapter 7 –Physical Layer and Transmission Media (Bandwidth Utilization).)
- 54.** Differentiate between Frequency Division Multiplexing (FDM) and Wavelength Division Multiplexing (WDM). CO5 (3.00)
(Theory) (Understanding(U)) (Medium) (Ref : Book) (Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Chapter 7 –Physical Layer and Transmission Media (Bandwidth Utilization).)